Lab 01 – Week 01

Identifying Super Keys, Candidate Keys and Primary key from Database Tables

Problem Statement:

Suppose, you are hired as a junior database developer in an XYZ software house. The software house gets a new project to develop the database of an ABC university. As a part of database development team, you are assigned the following tasks to perform.

Task 1:

Create a table named "Student" which should have the following attributes:

Roll Number, Student Name, Father Name, CNIC#, Mobile#, Degree Program, CGPA.

Roll Number	Student Name	Father Name	CNIC#	Mobile#	Degree Program	CGPA

Task 2:

Identify Super Keys, and Candidate Keys for the given table "Student".

Super Key: {Roll Number}, {CNIC#}, {Roll Number, Student Name}, {CNIC#, Student Name}, {Roll Number, Student Name, Father Name, CNIC#, Mobile#, Degree Program, CGPA}

Candidate Keys: {Roll Number}, {CNIC#}, {Mobile#}

Task 3:

Create another table named "Employee" with following attributes:

Emp ID, Name, CNIC, Salary, Designation, Email ID.

Emp ID	Name	CNIC	Salary	Designation,	Email ID

Task 4:

Identify Candidate Keys and Primary Key for the given table "Employee".

Candidate Keys: {Emp ID}, {CNIC}, {Email ID}

Primary Key: {Emp ID}

Good Luck!

Lab 02 – Week 02

Identifying Surrogated Keys and Foreign key from Database Tables

Problem Statement:

Suppose, you are hired as a junior database developer in an Abbot's solution. The software house gets a new project to develop the database for a new university. As a part of database development team, you are assigned the following tasks to perform.

Task 1:

Create a table named "Student" which should have the following attributes:

Roll Number, Student Name, Father Name, CNIC#, Mobile#, Degree Program, CGPA.

Roll Number	Student Name	Father Name	CNIC#,	Mobile#,	Degree Program	CGPA

Task 2:

Identify Surrogated Keys for the given table "Student".

Surrogated Keys: {Application Number}

Task 3:

Create another table named "Employee" with following attributes:

Emp ID, Name, CNIC, Salary, Designation, Email ID.

Emp ID	Name	CNIC	Salary	Designation	Email ID

Create another table named "Department" with following attributes:

Dpt ID, Dpt Name, Emp ID, Tel No

Dpt ID	Dpt Name	Emp ID	Tel No

Task 4:

Identify Primary key and Foreign Key for the given tables “Employee” and “Department”.

Employee Table:

Primary key: {Emp ID}

Foreign Key: {}

Department Table:

Primary key: {Dpt ID}

Foreign Key: {Emp ID}

Good Luck!

Lab 03 – Week 03

Problem Statement:

The ABC Company aims to develop a project management software which enables its users to manage and track projects, deadlines, schedules, and other workflow processes.

Each user has to perform a specific role. User can create tasks along with discussion threads. Users can post messages in Discussion Threads. User can also able to edit tasks and set deadlines. A complete progress can be viewed for the specific task.

You're required to perform the following tasks according to above scenario.

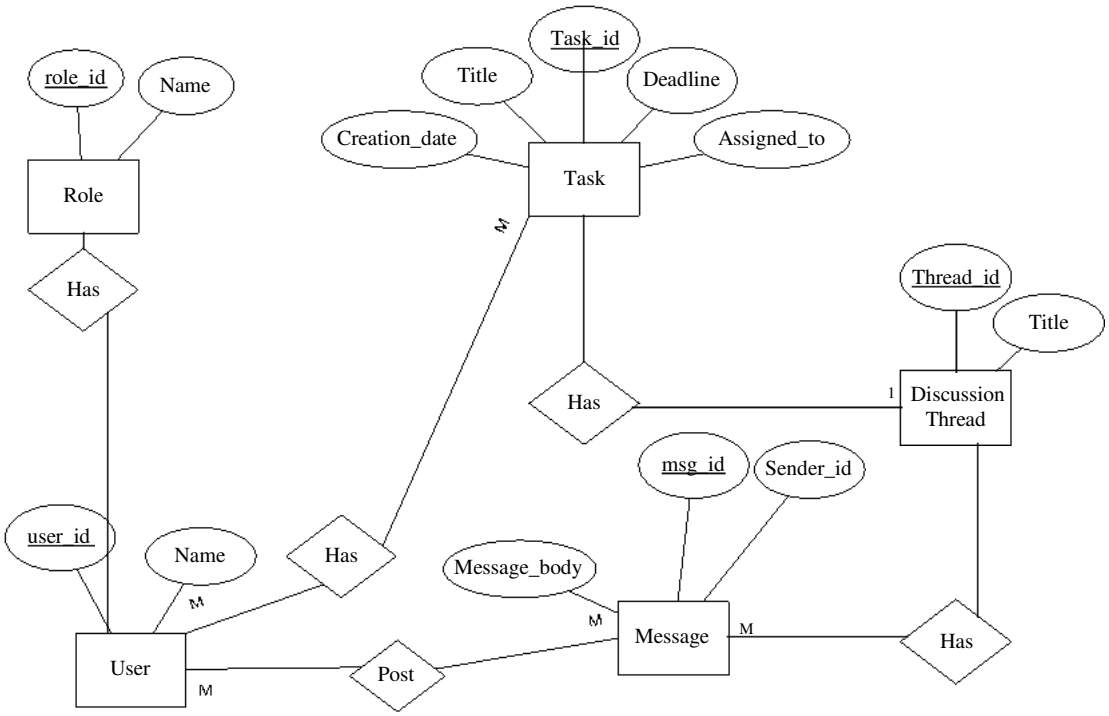
Task 1

Identify Entities, their attributes, relationships among entities in above scenario

Entities	Attributes
Role	role_id, name
User	user_id, name
Task	Task_id, title, creation_date, deadline, assigned_to
Discussion Thread	Thread_id, title
Messages	msg_id, message_body, sender_id

Task 2

Draw Entity Relationship Diagram (ERD) for it.


role_id
Name
Role
Has
Task_id
Title
Deadline
Creation_date
Assigned_to
Task
M
Has
Thread_id
Title
1
Discussion Thread
msg_id
Sender_id
user_id
Name
Has
Message_body
M
Message
M
Has
M
User
Post

Lab 04: Installation of Oracle 11g

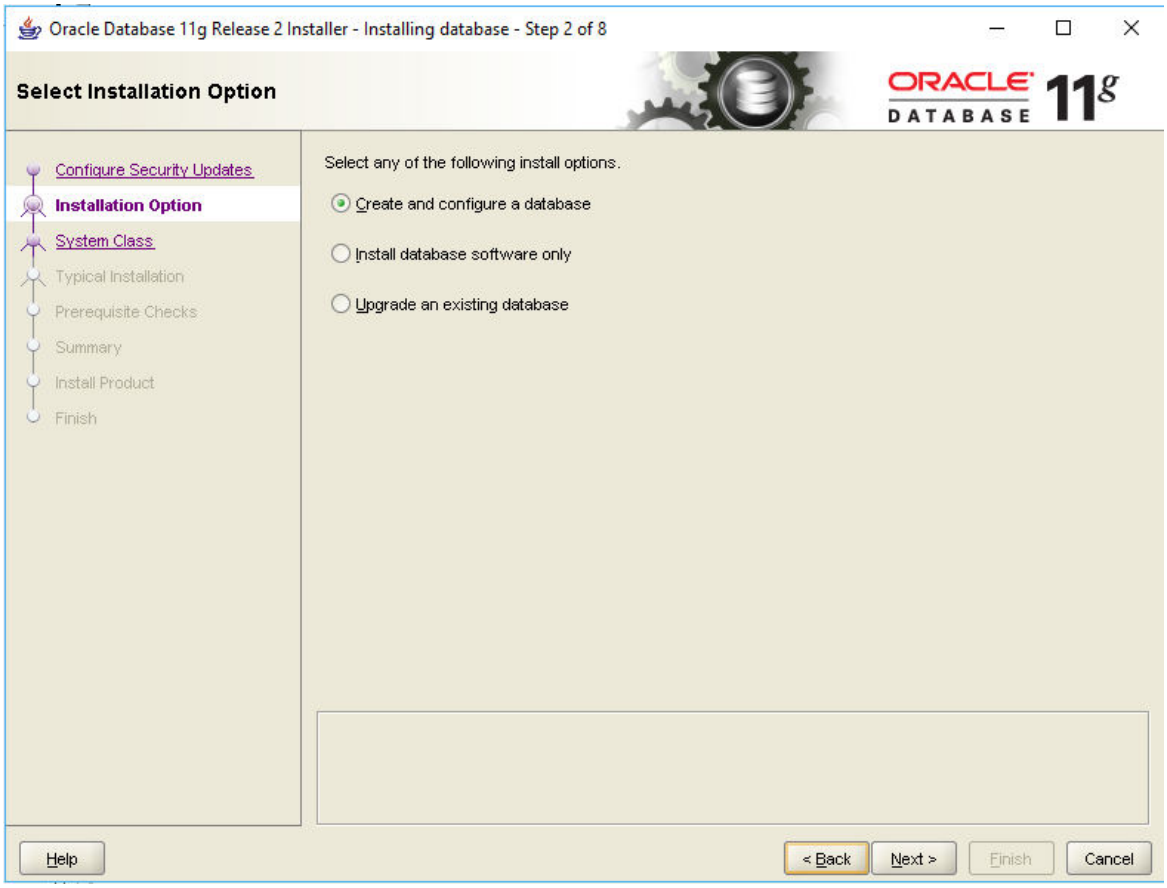
Task 1: You are required to install Oracle 11g in your system.

Installation Steps:

Before Installing Oracle, you must install complete OS

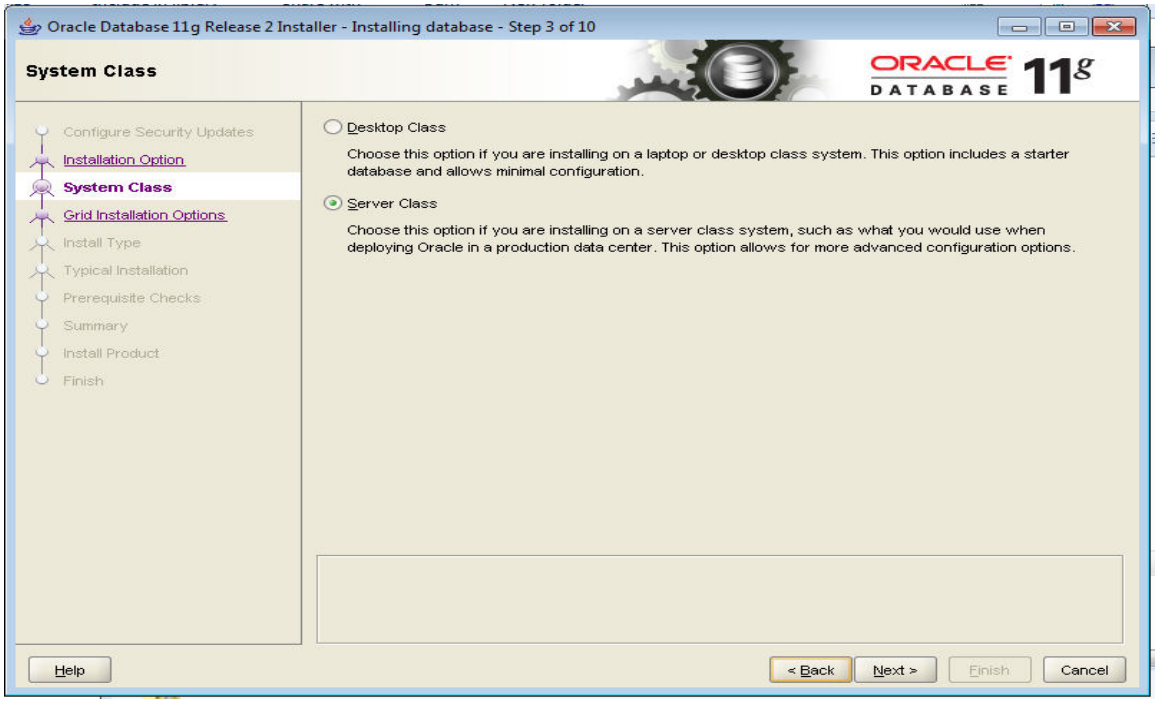
windows with all patches and activations. You may get this pop up in case if you have oracle previous

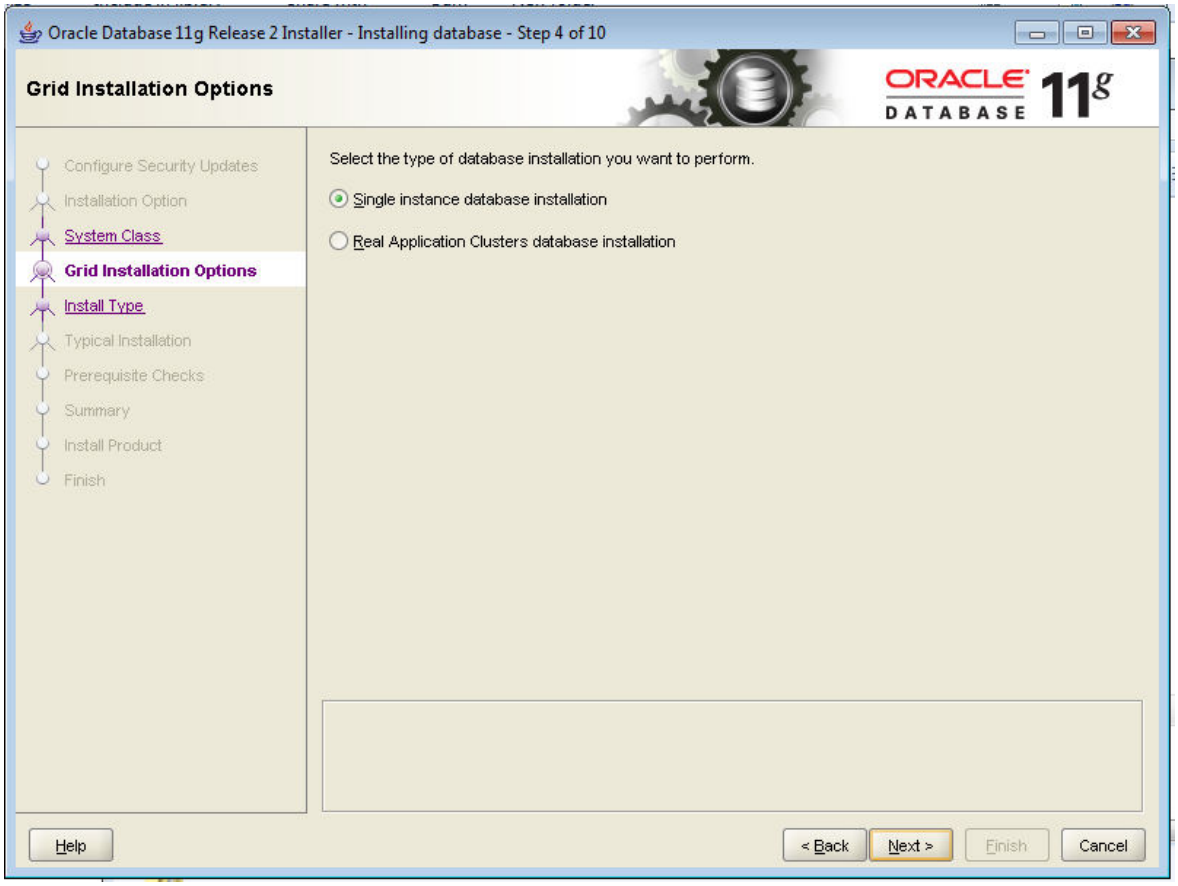
Installation, other proceed with next choice.

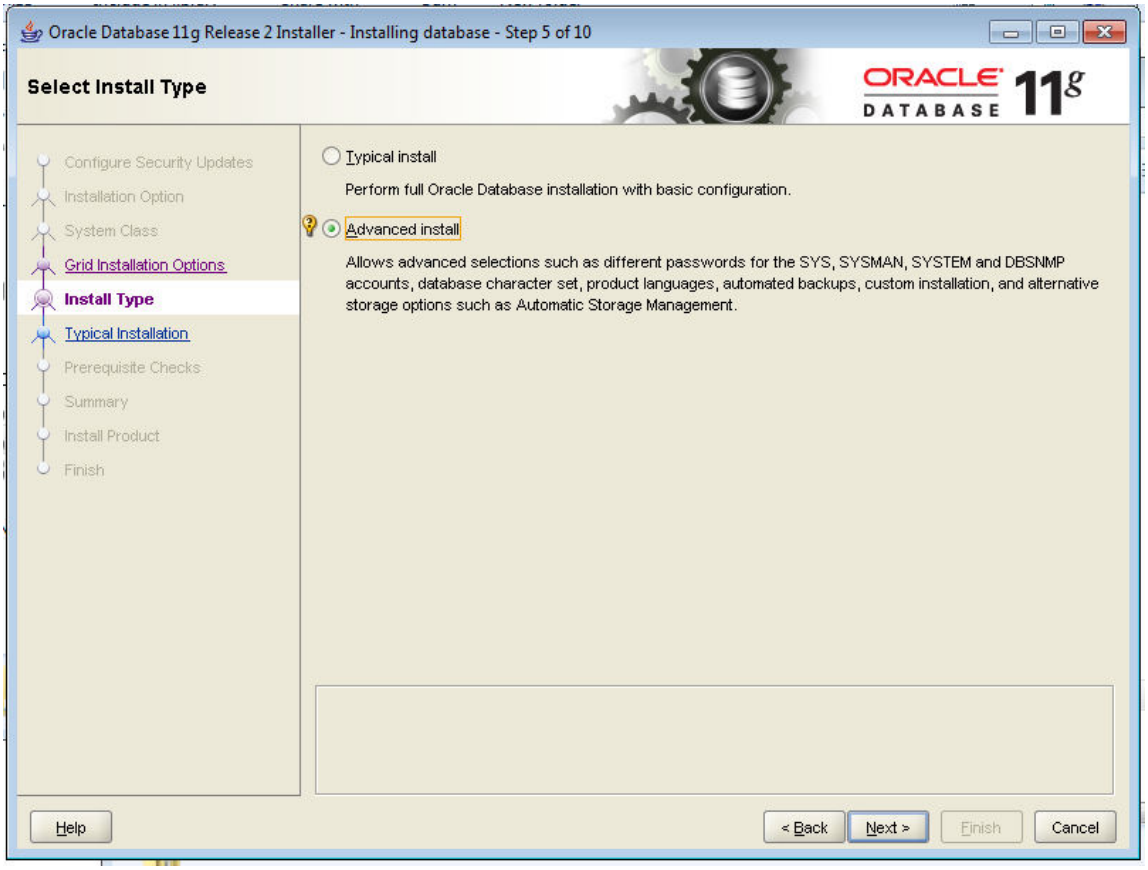


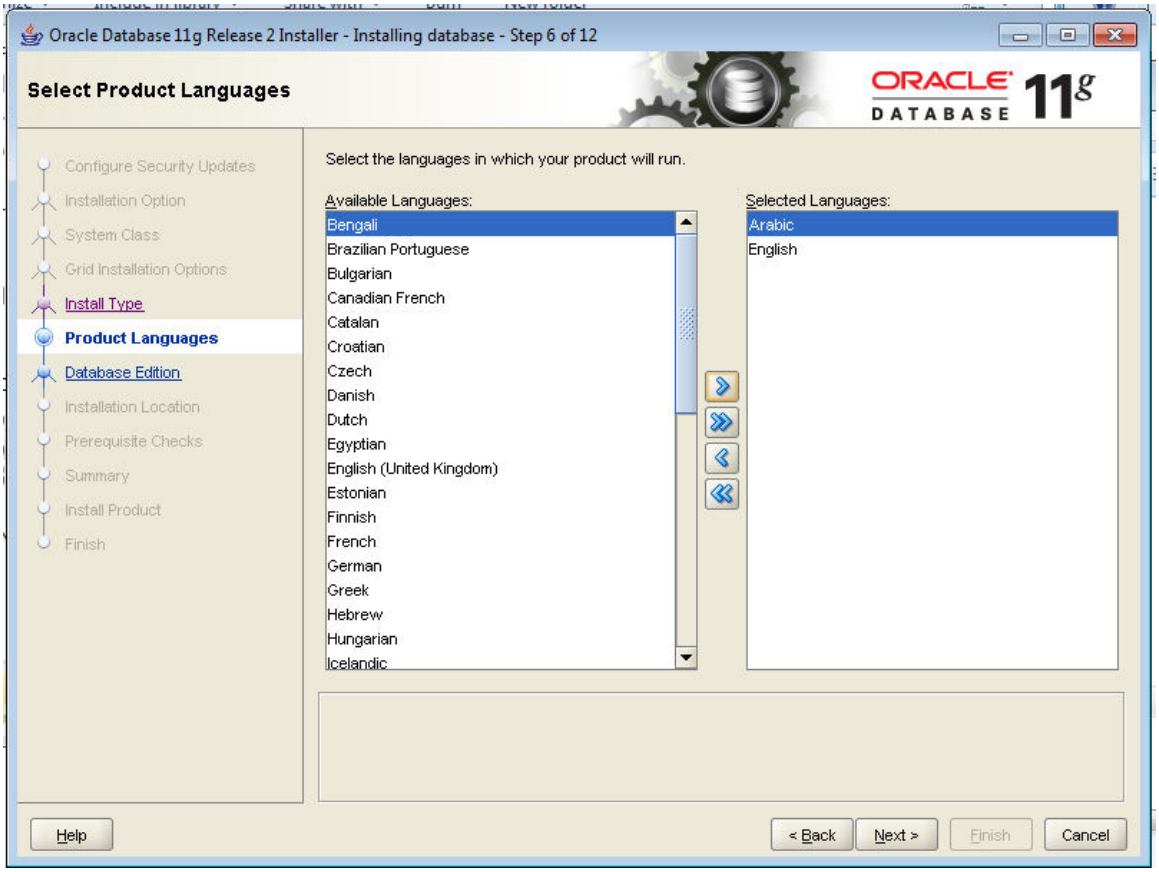
Before Installing Oracle, you must install complete OS windows with all patches and activations

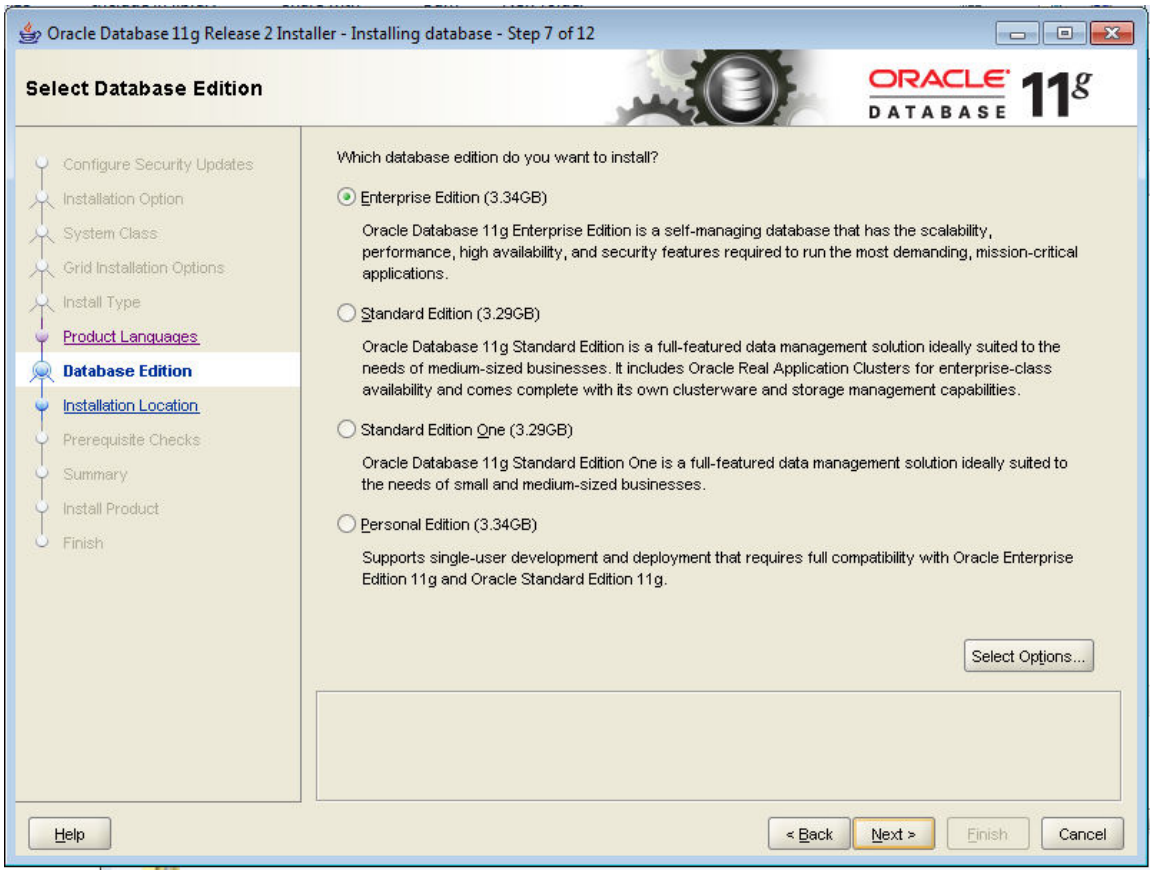
Run SETUP.EXE to start Oracle Universal Installer.

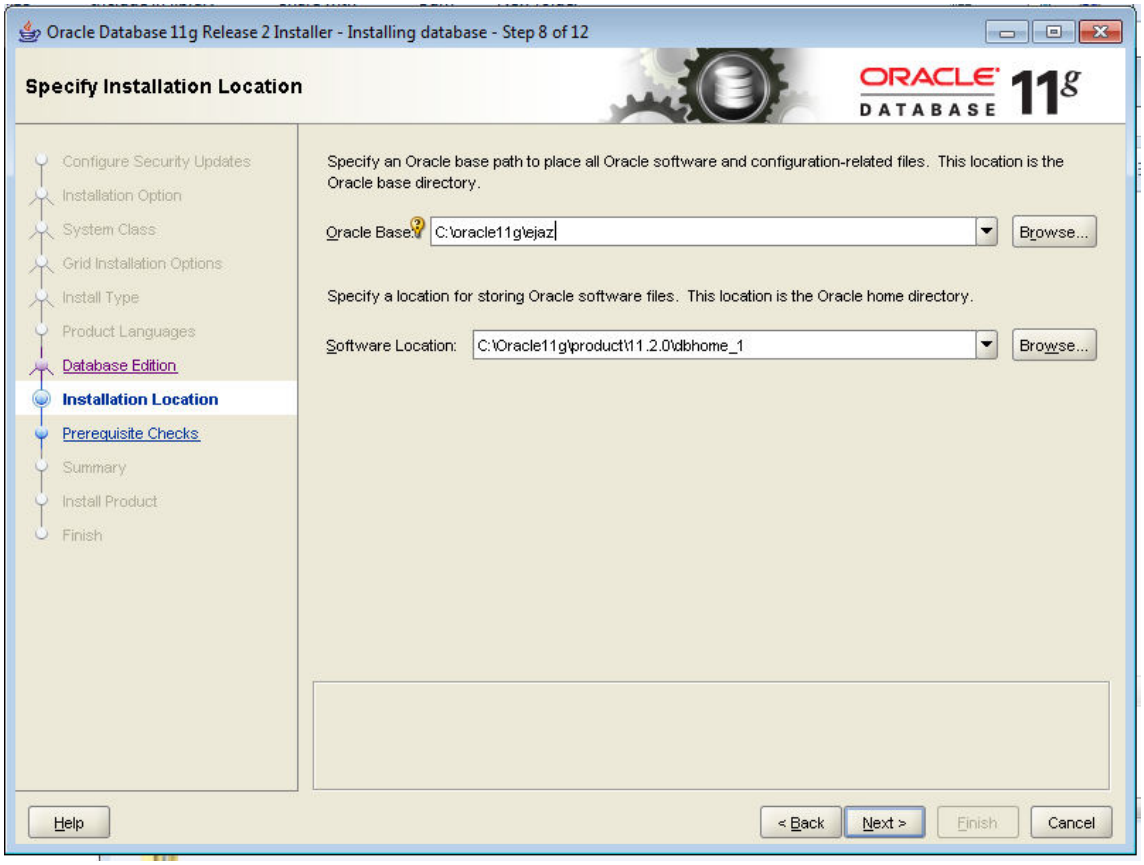

Oracle Database 11g Release 2 Installer - Installing database - Step 3 of 10
System Class
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Typical Installation
Prerequisite Checks
Summary
Install Product
Finish
Desktop Class
Choose this option if you are installing on a laptop or desktop class system. This option includes a starter database and allows minimal configuration.
Server Class
Choose this option if you are installing on a server class system, such as what you would use when deploying Oracle in a production data center. This option allows for more advanced configuration options.
Help
< Back
Next >
Finish
Cancel

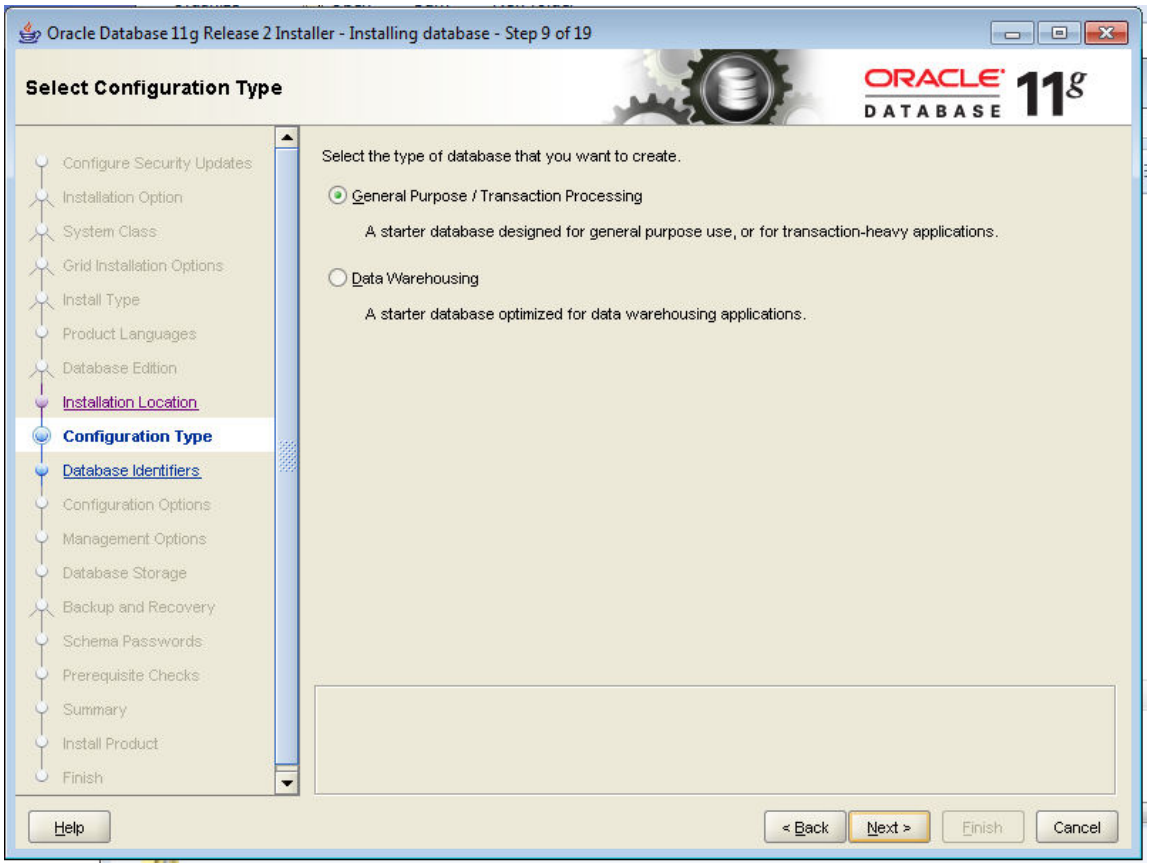

Oracle Database 11g Release 2 Installer - Installing database - Step 4 of 10
Grid Installation Options
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Typical Installation
Prerequisite Checks
Summary
Install Product
Finish
Select the type of database installation you want to perform.
Single instance database installation
Real Application Clusters database installation
Help
< Back
Next >
Finish
Cancel

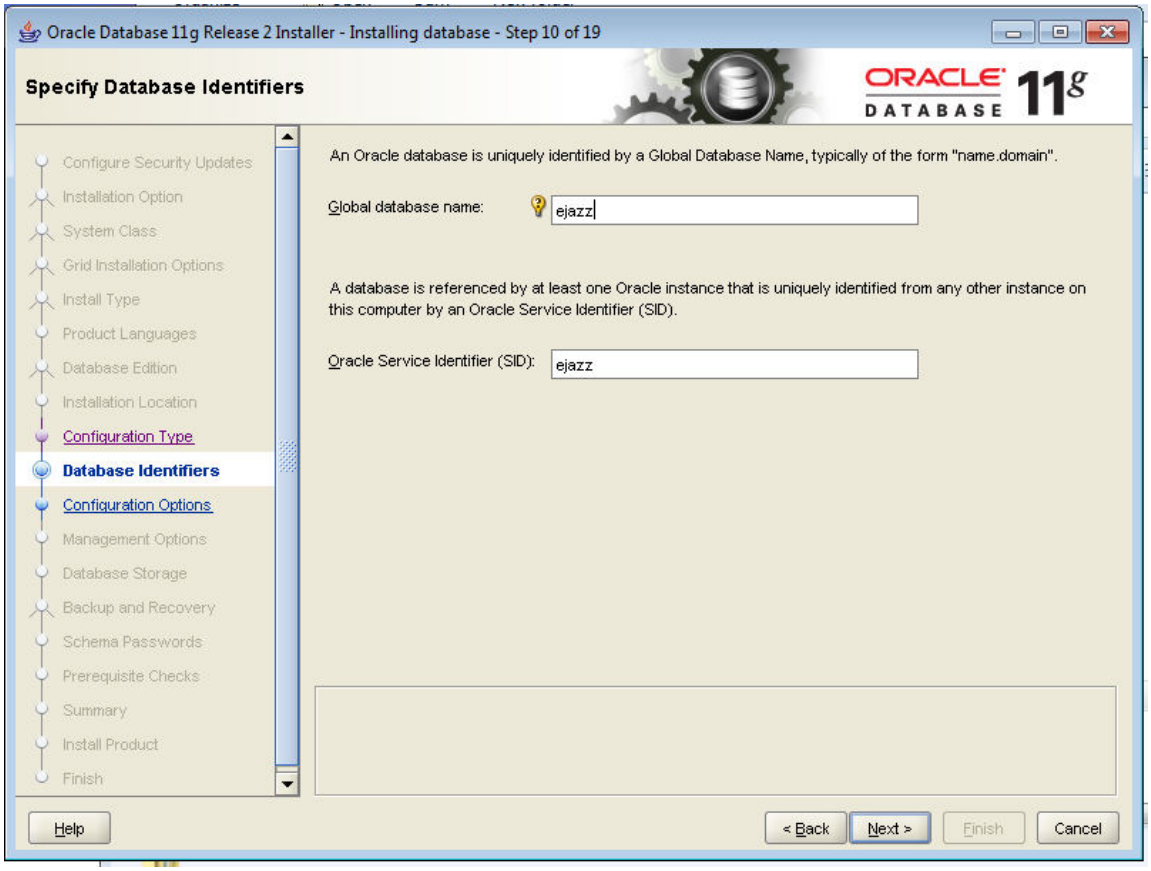

Oracle Database 11g Release 2 Installer - Installing database - Step 5 of 10
Select Install Type
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Typical Installation
Prerequisite Checks
Summary
Install Product
Finish
Typical install
Perform full Oracle Database installation with basic configuration.
Advanced install
Allows advanced selections such as different passwords for the SYS, SYSMAN, SYSTEM and DBSNMP accounts, database character set, product languages, automated backups, custom installation, and alternative storage options such as Automatic Storage Management.
Help
< Back
Next >
Finish
Cancel

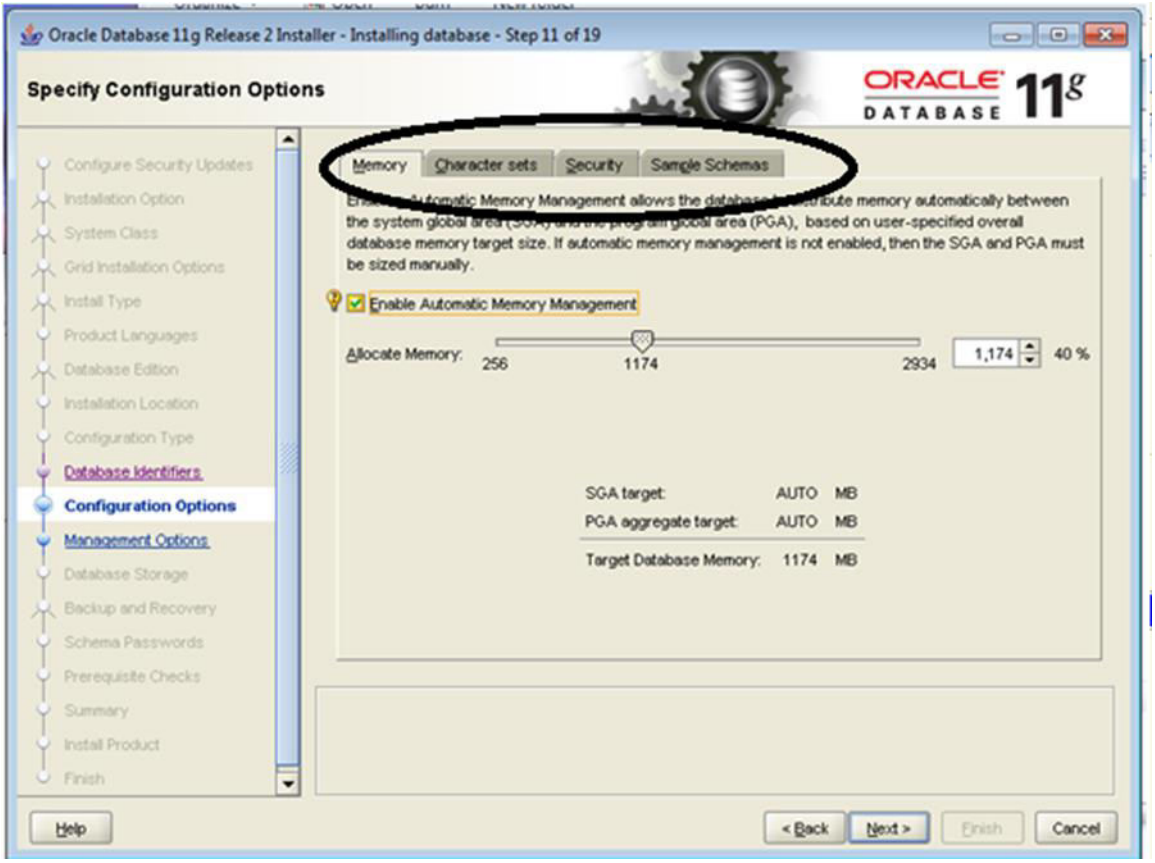

Oracle Database 11g Release 2 Installer - Installing database - Step 6 of 12
Select Product Languages
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Prerequisite Checks
Summary
Install Product
Finish
Select the languages in which your product will run.
Available Languages:
Bengali
Brazilian Portuguese
Bulgarian
Canadian French
Catalan
Croatian
Czech
Danish
Dutch
Egyptian
English (United Kingdom)
Estonian
Finnish
French
German
Greek
Hebrew
Hungarian
Icelandic
Selected Languages:
Arabic
English
Help
< Back
Next >
Finish
Cancel

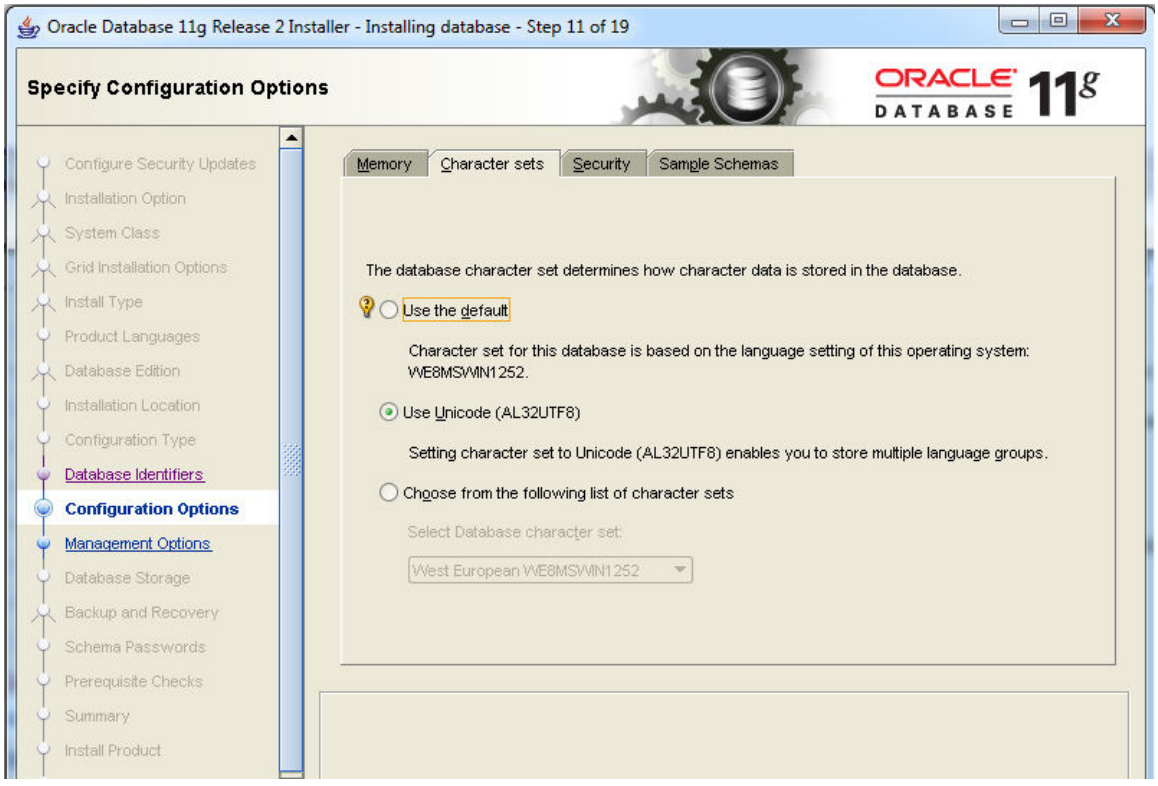

Oracle Database 11g Release 2 Installer - Installing database - Step 7 of 12
Select Database Edition
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Prerequisite Checks
Summary
Install Product
Finish
Which database edition do you want to install?
Enterprise Edition (3.34GB)
Oracle Database 11g Enterprise Edition is a self-managing database that has the scalability, performance, high availability, and security features required to run the most demanding, mission-critical applications.
Standard Edition (3.29GB)
Oracle Database 11g Standard Edition is a full-featured data management solution ideally suited to the needs of medium-sized businesses. It includes Oracle Real Application Clusters for enterprise-class availability and comes complete with its own clusterware and storage management capabilities.
Standard Edition One (3.29GB)
Oracle Database 11g Standard Edition One is a full-featured data management solution ideally suited to the needs of small and medium-sized businesses.
Personal Edition (3.34GB)
Supports single-user development and deployment that requires full compatibility with Oracle Enterprise Edition 11g and Oracle Standard Edition 11g.
Select Options...
Help
< Back
Next >
Finish
Cancel

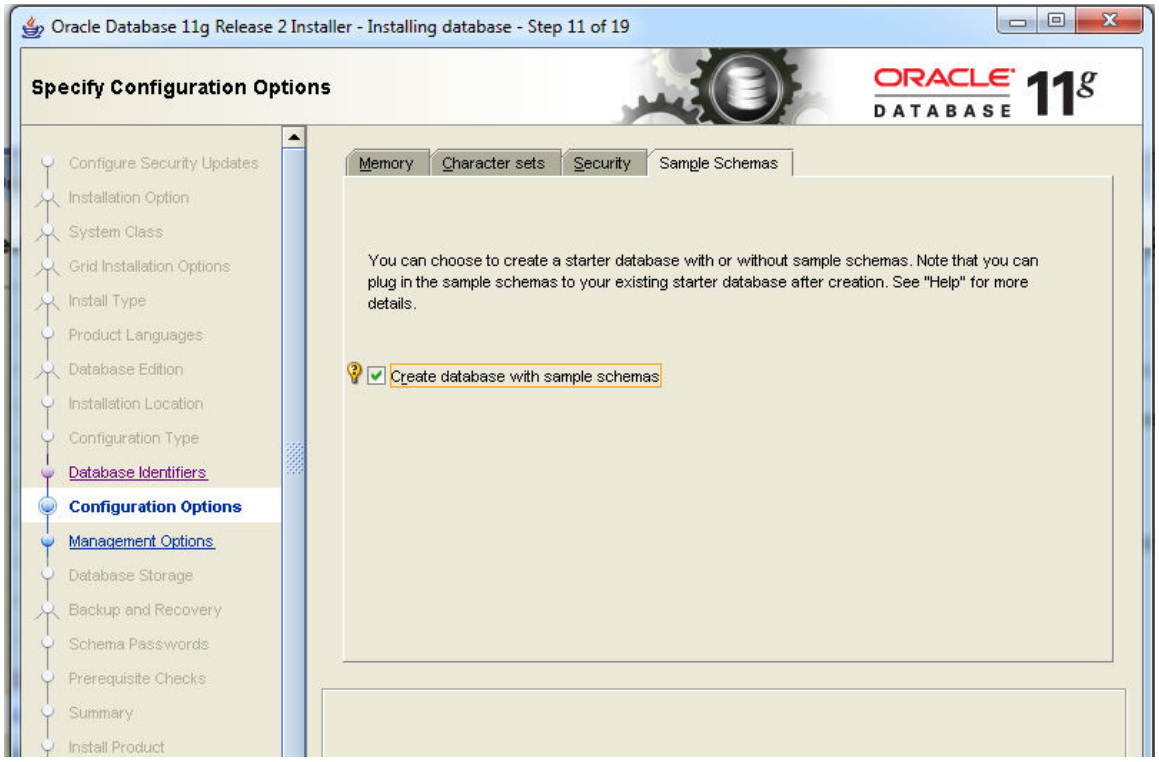

Oracle Database 11g Release 2 Installer - Installing database - Step 8 of 12
Specify Installation Location
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Prerequisite Checks
Summary
Install Product
Finish
Specify an Oracle base path to place all Oracle software and configuration-related files. This location is the Oracle base directory.
Oracle Base:
C:\oracle11g\ejaz
Browse...
Specify a location for storing Oracle software files. This location is the Oracle home directory.
Software Location:
C:\Oracle11g\product\11.2.0\dbhome_1
Browse...
Help
< Back
Next >
Finish
Cancel

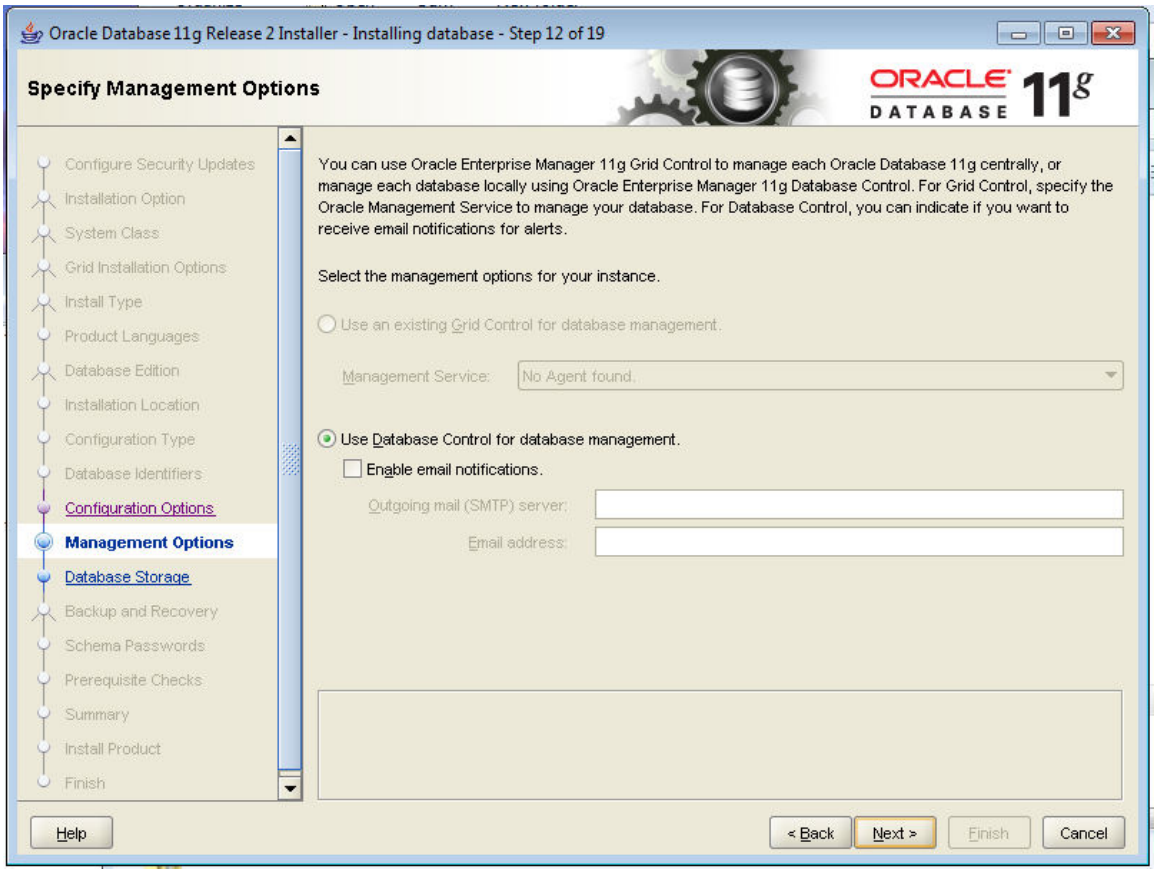

Oracle Database 11g Release 2 Installer - Installing database - Step 9 of 19
Select Configuration Type
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish
Select the type of database that you want to create.
General Purpose / Transaction Processing
A starter database designed for general purpose use, or for transaction-heavy applications.
Data Warehousing
A starter database optimized for data warehousing applications.
Help
< Back
Next >
Finish
Cancel

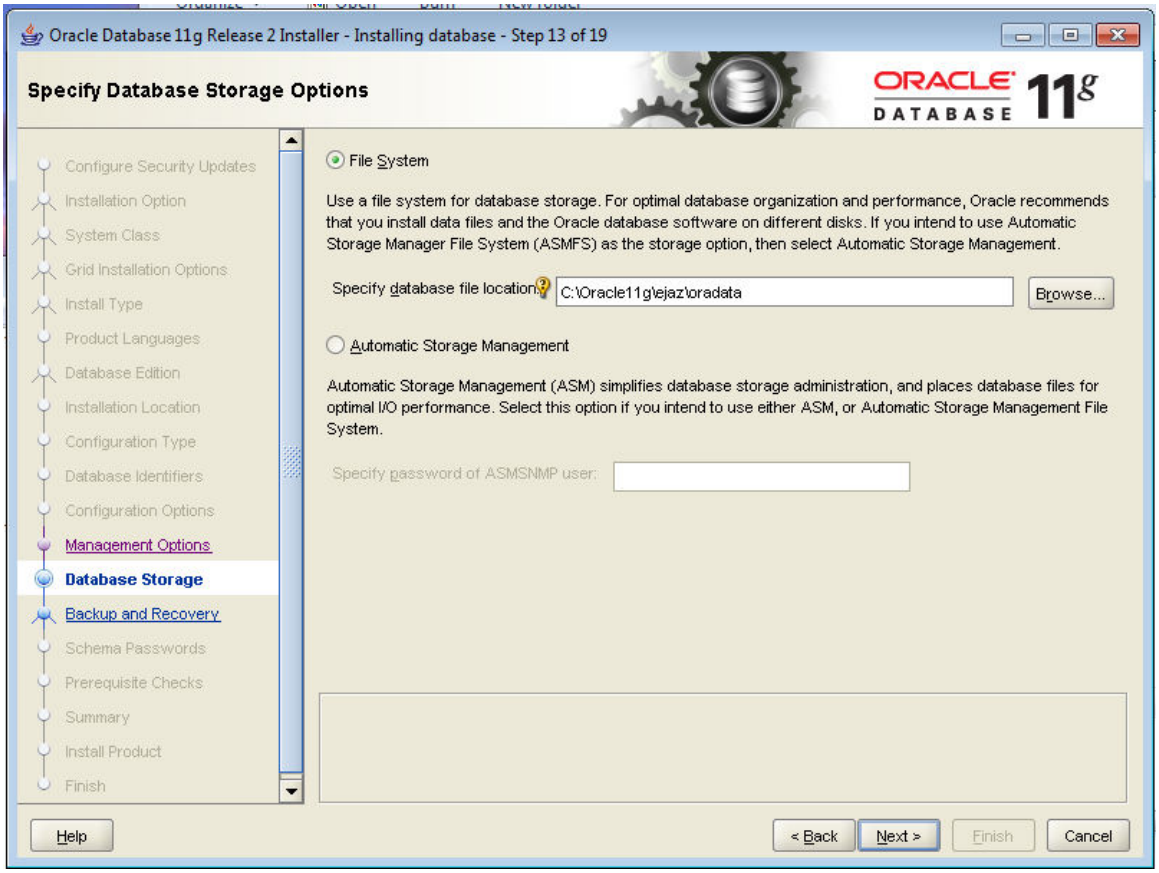

Oracle Database 11g Release 2 Installer - Installing database - Step 10 of 19
Specify Database Identifiers
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish
An Oracle database is uniquely identified by a Global Database Name, typically of the form "name.domain".
Global database name:
ejazz
A database is referenced by at least one Oracle instance that is uniquely identified from any other instance on this computer by an Oracle Service Identifier (SID).
Oracle Service Identifier (SID):
ejazz
Help
< Back
Next >
Finish
Cancel

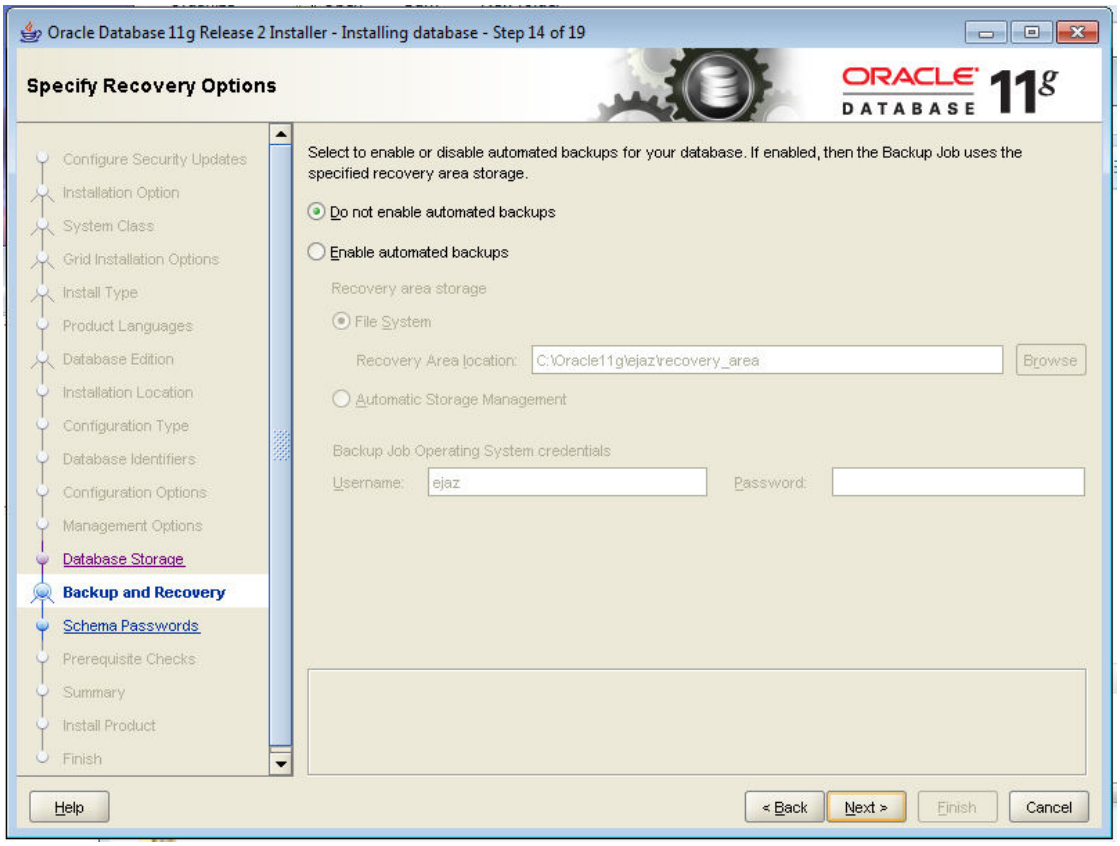

Oracle Database 11g Release 2 Installer - Installing database - Step 11 of 19
Specify Configuration Options
ORACLE 11g DATABASE
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish
Memory
Character sets
Security
Sample Schemas
Automatic Memory Management allows the database to distribute memory automatically between the system global area (SGA) and the program global area (PGA), based on user-specified overall database memory target size. If automatic memory management is not enabled, then the SGA and PGA must be sized manually.
Enable Automatic Memory Management
Allocate Memory:
256
1174
2934
1,174
40 %
SGA target: AUTO MB
PGA aggregate target: AUTO MB
Target Database Memory: 1174 MB
Help
< Back
Next >
Finish
Cancel

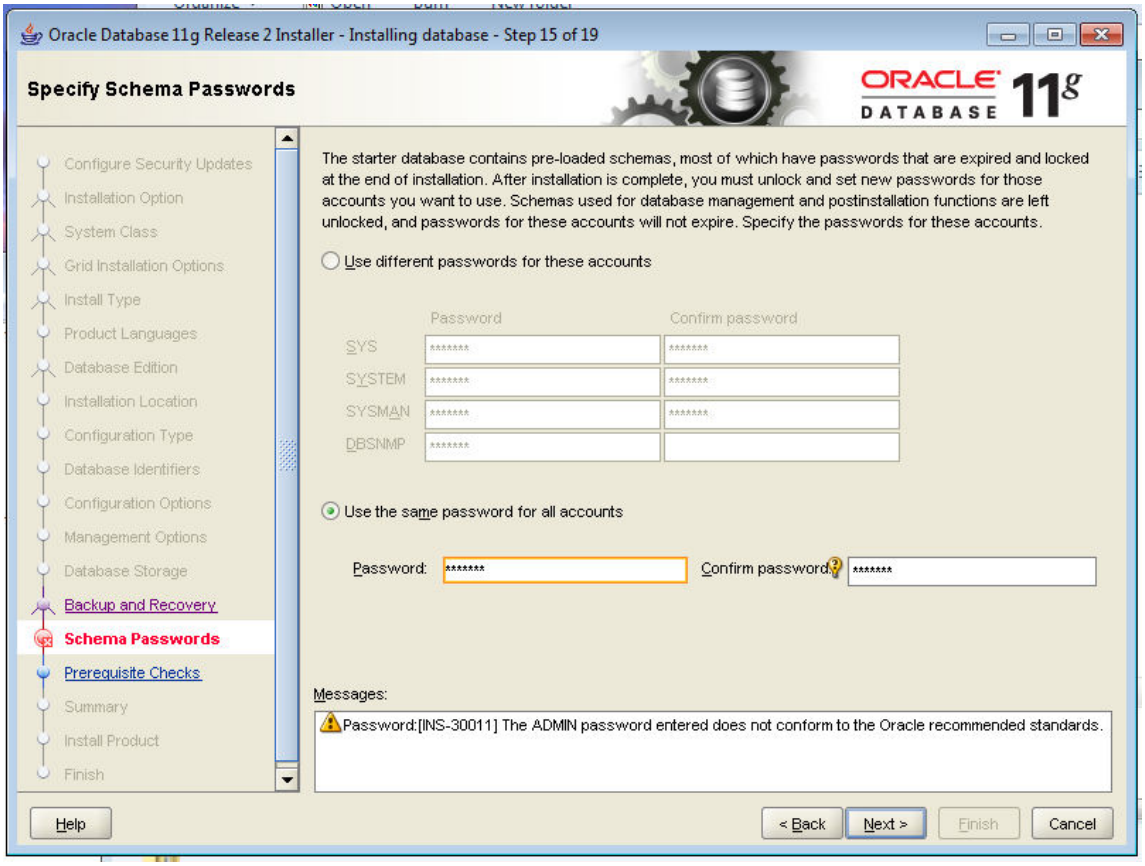

Oracle Database 11g Release 2 Installer - Installing database - Step 11 of 19
Specify Configuration Options
ORACLE 11g DATABASE
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Memory
Character sets
Security
Sample Schemas
The database character set determines how character data is stored in the database.
Use the default
Character set for this database is based on the language setting of this operating system: WE8MSWIN1252.
Use Unicode (AL32UTF8)
Setting character set to Unicode (AL32UTF8) enables you to store multiple language groups.
Choose from the following list of character sets
Select Database character set:
West European WE8MSWIN1252

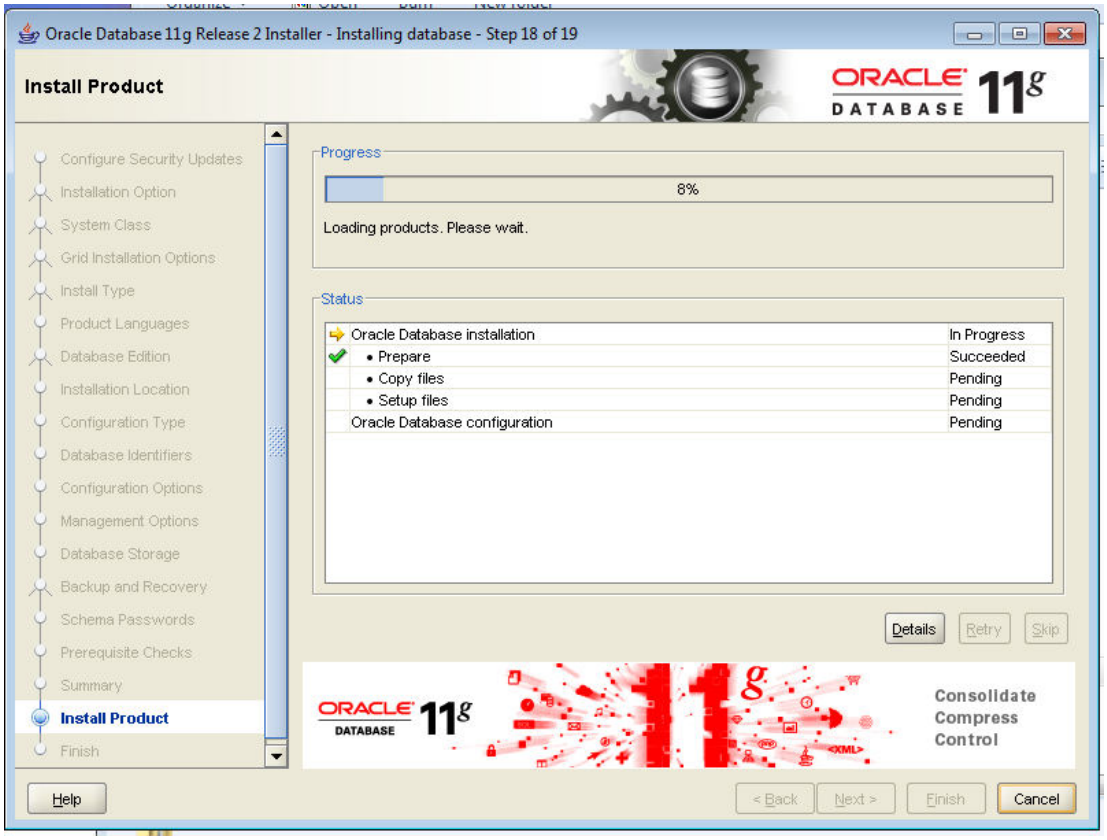

Oracle Database 11g Release 2 Installer - Installing database - Step 11 of 19
Specify Configuration Options
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Memory
Character sets
Security
Sample Schemas
You can choose to create a starter database with or without sample schemas. Note that you can plug in the sample schemas to your existing starter database after creation. See "Help" for more details.
Create database with sample schemas


Oracle Database 11g Release 2 Installer - Installing database - Step 12 of 19
Specify Management Options
ORACLE DATABASE 11g
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish
You can use Oracle Enterprise Manager 11g Grid Control to manage each Oracle Database 11g centrally, or manage each database locally using Oracle Enterprise Manager 11g Database Control. For Grid Control, specify the Oracle Management Service to manage your database. For Database Control, you can indicate if you want to receive email notifications for alerts.
Select the management options for your instance.
Use an existing Grid Control for database management.
Management Service:
No Agent found.
Use Database Control for database management.
Enable email notifications.
Outgoing mail (SMTP) server:
Email address:
Help
< Back
Next >
Finish
Cancel


Oracle Database 11g Release 2 Installer - Installing database - Step 13 of 19
Specify Database Storage Options
ORACLE 11g DATABASE
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish
File System
Use a file system for database storage. For optimal database organization and performance, Oracle recommends that you install data files and the Oracle database software on different disks. If you intend to use Automatic Storage Manager File System (ASMFS) as the storage option, then select Automatic Storage Management.
Specify database file location:
C:\Oracle11g\ejaz\oradata
Browse...
Automatic Storage Management
Automatic Storage Management (ASM) simplifies database storage administration, and places database files for optimal I/O performance. Select this option if you intend to use either ASM, or Automatic Storage Management File System.
Specify password of ASMSNMP user:
Help
< Back
Next >
Finish
Cancel

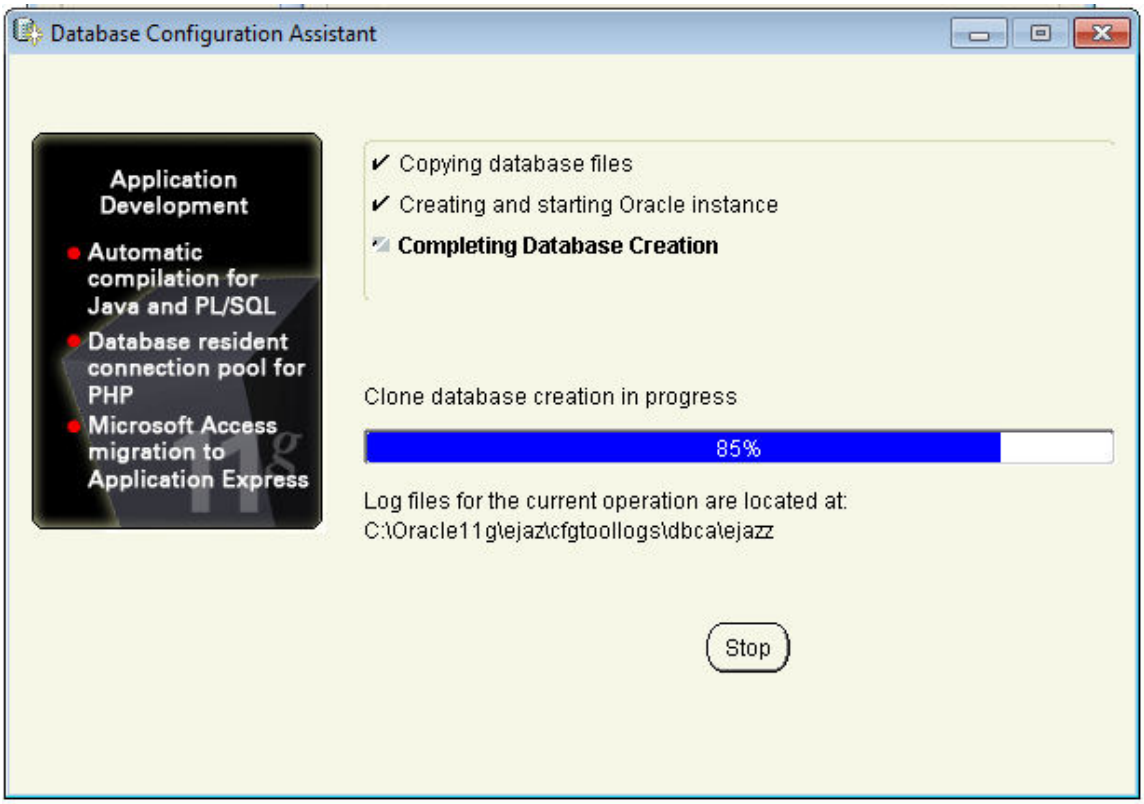

Oracle Database 11g Release 2 Installer - Installing database - Step 14 of 19
Specify Recovery Options
ORACLE 11g DATABASE
Configure Security Updates
Installation Option
System Class
Grid Installation Options
Install Type
Product Languages
Database Edition
Installation Location
Configuration Type
Database Identifiers
Configuration Options
Management Options
Database Storage
Backup and Recovery
Schema Passwords
Prerequisite Checks
Summary
Install Product
Finish
Select to enable or disable automated backups for your database. If enabled, then the Backup Job uses the specified recovery area storage.
Do not enable automated backups
Enable automated backups
Recovery area storage
File System
Recovery Area location:
C:\Oracle11g\ejaz\recovery_area
Browse
Automatic Storage Management
Backup Job Operating System credentials
Username:
ejaz
Password:
Help
< Back
Next >
Finish
Cancel

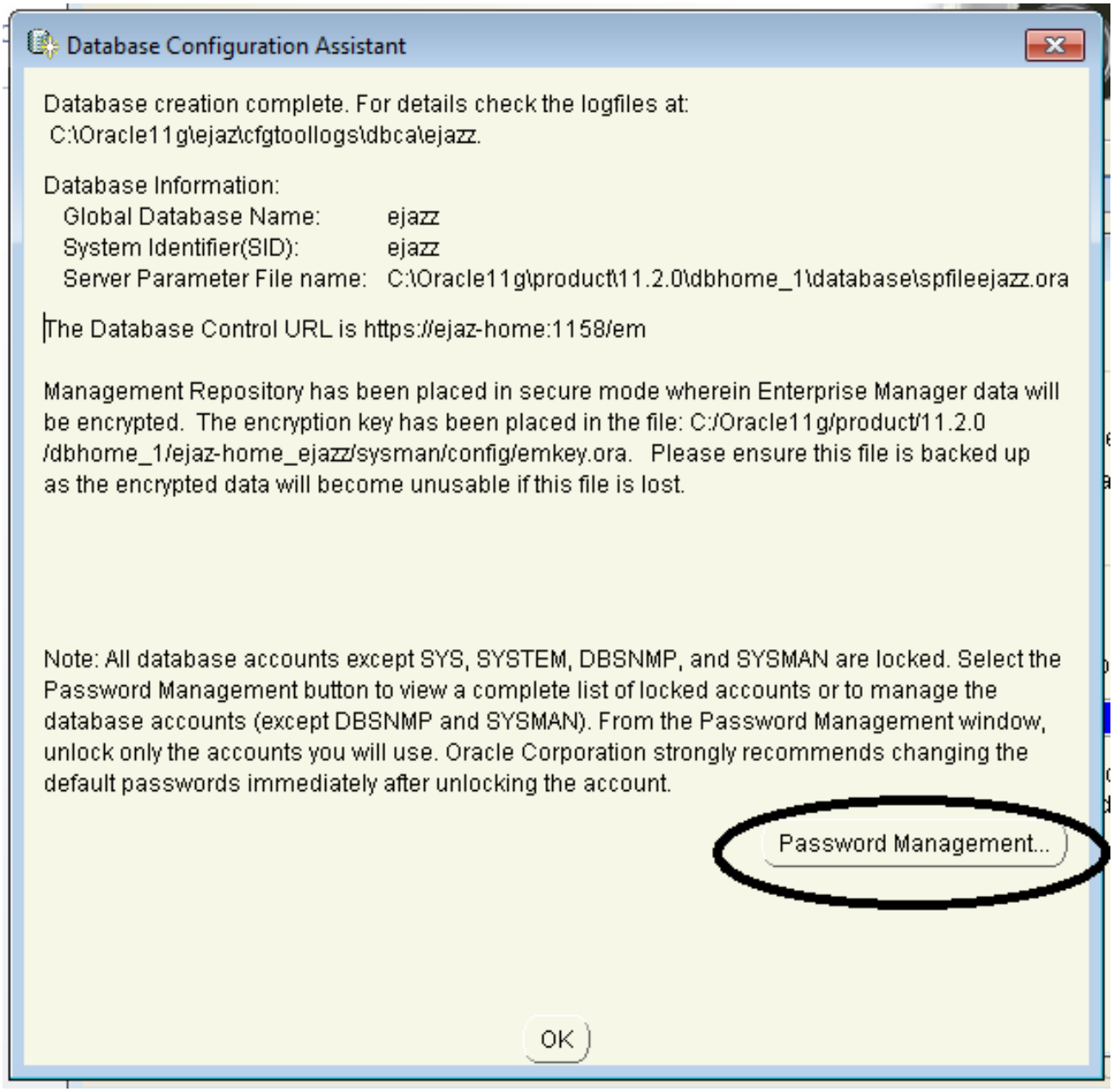




In case if any process failed, press Retry button.

If NetConfig error displays then press continue





Login with Oracle Database Manager

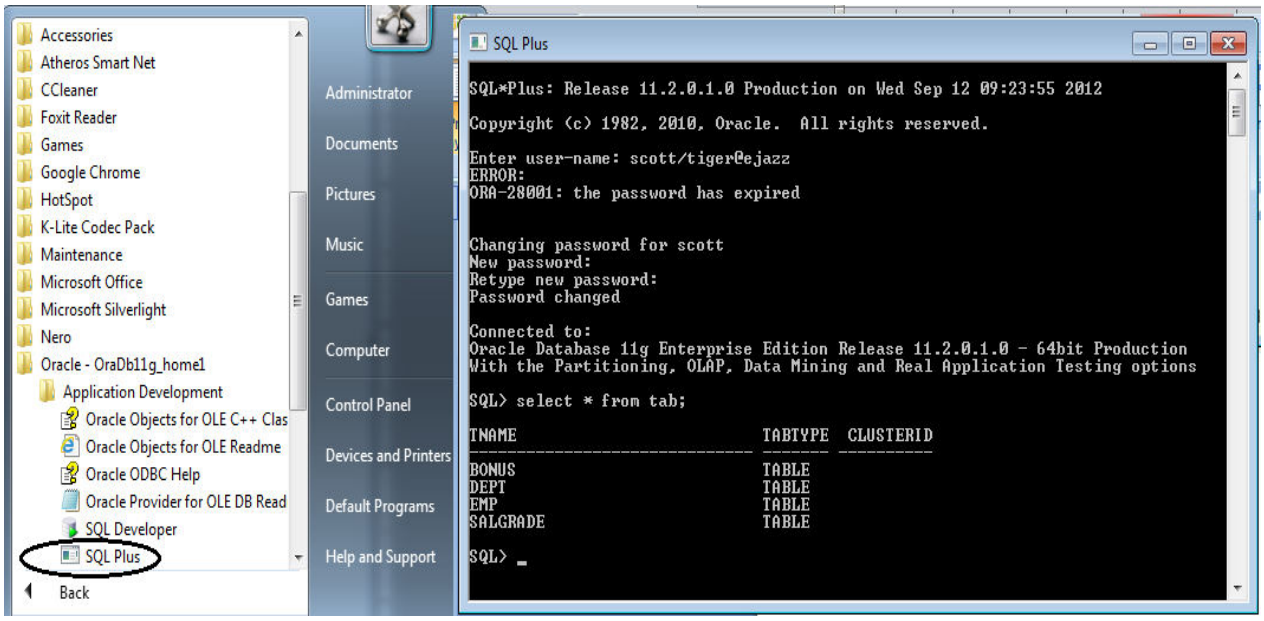
Not secure | haider.mshome.net:5500/em/console/logon/logon

ORACLE Enterprise Manager 11g
Database Control

Login

* User Name: sys
* Password: ••••••••
Connect As: SYSDBA
Login

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Lab 05: Redo Log Files

Task 1:

Write an SQL statement to create a new group of redo log files. The redo log files names should be respectively log1b.rdo, logbc.rdo and their size should be 100 MB.

```
ALTER DATABASE
ADD LOGFILE
('/oracle/dbs/log1b.rdo',
 '/oracle/dbs/log2b.rdo')
SIZE 100M;
```

Task 2:

Write an SQL statement to create a new group of redo log files with group number. The redo log files names should be respectively log1c.rdo, log2c.rdo. Their size, BLOCK SIZE and group number should be 100 MB, 512, 10 respectively.

```
ALTER DATABASE
ADD LOGFILE GROUP 10 ('/oracle/dbs/log1c.rdo', '/oracle/dbs/log2c.rdo')
SIZE 100M BLOCKSIZE 512;
```

Task 3:

Write a SQL following statement that add a new redo log member (log2b.rdo) to redo log group number 3.

```
ALTER DATABASE
ADD LOGFILE MEMBER '/oracle/dbs/log2b.rdo' TO GROUP 3;
```

Task4:

Write an SQL statement to drop the redo log group number 2?

```
ALTER DATABASE DROP LOGFILE GROUP 2;
```

Lab 06-Week 06

Topic: Tablespaces

Task 1:

Write an SQL statement to create a new tablespace named **NEWTB** and its size should be **50 MB**.

```
CREATE TABLESPACE NEWTB
datafile 'C:\Oracle11g\ordata\wrk01.dbf'
size 50M online;
```

Task 2:

Write an SQL statement to create locally-Managed Tablespace named **user1** and its size should be **10 MB**?

```
CREATE TABLESPACE user1
datafile 'C:\Ora11g\ordata\usr01.dbf'
size 10M
EXTENT MANAGEMENT LOCAL UNIFORM SIZE 128K;
```

Task 3:

Write a SQL statement to create the undo tablespace name undo10 and its size should be 20MB.

```
CREATE UNDO TABLESPACE undo10
datafile 'C:\Oracle11g\ordata\undo01.dbf'
size 20M;
```

Lab 07 - Week 07

Topic: Creating and Alter table

Task 1:

Write SQL command to create the **Employee** table having the following columns and their constraints. You are required to choose the appropriate data type for each column by your own.

Column Name	Data Type	Constraint
EMP_No	-	NOT NULL
EMP_Name	-	NOT NULL
EMP_Age	-	DEFAULT 0
EMP_City	-	
Salary	-	NOT NULL

```
CREATE TABLE Employee (
EMP_No          INT NOT NULL,
EMP_Name        Varchar2(20) NOT NULL,
EMP_Age         Number(5) DEFAULT 0,
EMP_City        Varchar2(40)

Salary          Number(6) NOT NULL

);
```

Task 2:

Consider the following Employee table:

EMP_No	EMP_Name	EMP_Age	EMP _City	Salary

You are required to write PL/SQL code for an ORACLE transaction on employee table (given above) containing the following SQL commands.

- INSERT (ADD) a new record having EMP_NO as “006” , EMP_Name as “ALI” , EMP_Age as “30” , EMP_City as “LHR” and Salary as “55000”.
- Delete “EMP_City” column from employee relation.

Solution:

```
DECLARE

BEGIN
```

```
        INSERT INTO Employee (EMP_NO, EMP_Name, EMP_Age, EMP_City, Salary)
VALUES (‘006’, ‘ALI’, ‘30’, ‘LHR’, ‘55000’);


        ALTER TABLE Employee
        Drop column ‘EMP_City’;
        COMMIT;
EXCEPTION
        WHEN OTHERS THEN
        ROLLBACK;
END;
/
```

Lab 08 - Week 08

Topic: Create & Alter Index

Task 1:

Write PL/SQL command to create an Index on the name column of give table of employee?

EMP_No	EMP_Name	EMP_Age	EMP_City	Salary

CREATE INDEX Emp_idx ON Employee (EMP_Name);

Task 2:

Write an PL/SQL command to rename the index called *EMP_idx* to *EMP_index_name*.

ALTER INDEX EMP_idx RENAME TO EMP_index_name;

Task 3:

Write SQL command to create an Index named ENAME_index on the EMP_Name column of above given table of employee. Tablespace should be used “users” and storage size is initial 20k and next is 20K.

CREATE INDEX ENAME_index ON Employee(EMP_Name)

TABLESPACE users

STORAGE (INITIAL 20K NEXT 20k);

Lab 09 - Week 09

Topic: PL/SQL Basics

Task 1:

Write an PL/SQL program to add two numbers and the sum of those two numbers is stored in a third variable and print the result on the screen.

```
SQL> Declare
  2  -- declare variable x, y and z of datatype number
  3  x number(5);
  4  y number(5);
  5  z number(7);
  6  Begin
  7  -- Here we Assigning 10 into x
  8  x:=10;
  9  -- Assigning 20 into x
 10  y:=20;
 11  -- Assigning sum of x and y into z
 12  z:=x+y;
 13  --Print the Result
 14  dbms_output.put_line('Sum is '||z);
 15  End;
 16  /
Sum is 30
```

Lab 10 - Week 10

Topic: PL/SQL Basics

Task 1:

Write an PL/SQL procedure that calculate factorial of a number using a function?

```
CREATE OR REPLACE FUNCTION fact(x number)
RETURN number IS
   f number;
BEGIN
   IF x=0 THEN
      f := 1;
   ELSE
      f := x * fact(x-1);
   END IF;
RETURN f;
END;
/

DECLARE
   num number;
   factorial number;
BEGIN
   num:= 6;
   factorial := fact(num);
   dbms_output.put_line(' Factorial '|| num || ' is ' || factorial);
END;
/
```

Lab11-Week 11

Topic: Users information using Data Dictionary Views

Task 1:

Write a PL/SQL command for listing all the tablespace quota?

```
Select * from DBA_TS_QUOTAS;
```

Task2:

Write a PL/SQL command for listing all profiles and assigned limit?

```
Select * from DBA_PROFILES ORDER BY PROFILE;
```

Lab 12 – WEEK 12

Topic: Managing and Creating User with Commands

TASK 1:

Write down the syntax for creating an oracle user with commands?

```
CREATE USER sidney
     IDENTIFIED BY out standing1
     DEFAULT TABLESPACE example
     QUOTA 10M ON example
     TEMPORARY TABLESPACE temp
     QUOTA 5M ON system
     PROFILE app_user
     PASSWORD EXPIRE;
```

```
CREATE USER global_user
    IDENTIFIED GLOBALLY AS 'CN=analyst
    DEFAULT TABLESPACE example
    QUOTA 5M ON example;
```

TASK 2:

Write a SQL command that display all tablespace quotas?

Answer: SELECT * FROM DBA_TS_QUOTAS;

TASK 3:

Write a SQL command that display user information like name, profile and account status from database.

SELECT USERNAME, PROFILE, ACCOUNT_STATUS FROM DBA_USERS;

Lab 13- week 13

Topic: Pre-defined Roles & Assigning Roles for DBA

Task 1:

Write the SQL command for altering the role of database administrator in oracle database?

ALTER ROLE role_name {NOT IDENTIFIED |IDENTIFIED

{By password | USING Package | EXTERNALLY | GLOBALLY}};

Task 2:

List down the pre-defined roles of database administrator in oracle database?

CONTENT

RESOURCE

DBA

EXP_FULL_DATABASE

IMP_FULL_DATABASE

DELETE_CATALOG_ROLE

EXECUTE_CATALOG_ROLE

SELECT_CATALOG_ROLE

Lab 14 – Week 14

Topic: SQL* Loader

Task 1:

Using SQL* Loader copy (load) data from the "Dept" table into the "Dept1" table without APPEND hint.

```
INSERT  INTO  Dept1

SELECT  *  FROM  Dept;

COMMIT;      -- commit command is necessary to load data
```

Task 2:

Using SQL* Loader copy (load) data from the "Product" table into the "Product1" table with APPEND hint.

```
INSERT  /*+  APPEND */  INTO  Product1

SELECT  *  FROM  Product;

COMMIT;      -- commit command is necessary to load data
```

Lab 15 (Solution)

Topics: Transaction Processing & Role of Transactions

Task 1:

Supposed you have the following table named "Employee". Write a Transaction (PL/SQL code) to update the salary of '4156897' employee number to 17900.

Employee

EMP_No	EMP_Title	Salary
4156897	Talha Akhtar	15500
3256984	Ali Khalid	21500

Solution:

```
BEGIN
UPDATE Employee
SET Salary=17900
WHERE EMP_NO=4156897;
DBMS_OUTPUT_LINE("Transaction updated successfully")
END;
```

Lab 16 (Solution)

Topics: Transaction Processing & Role of Transactions

Task 1:

Write SQL command to allocate (grant) SELECT, INSERT and UPDATE privileges on "Credit" table to all users.

Solution:

```
GRANT       SELECT, INSERT, UPDATE
ON          Credit
TO          PUBLIC;
```